

Acme Water Purifier W3

Service Manual

1. Product Overview

The Acme Water Purifier W3 is a 4-stage RO + UV countertop purifier with a 1.2L pressure tank. This manual covers the filter stages, UV lamp, and pump. Disconnect power and close the inlet valve before any service.

2. Components

- Stage 1 — sediment pre-filter (PP spun, 5 micron).
- Stage 2 — activated carbon block, removes chlorine and odor.
- Stage 3 — RO membrane (75 GPD), the core purification element.
- Stage 4 — UV sterilization lamp with ballast and flow sensor.
- Booster pump — raises inlet pressure for the RO membrane.
- Solenoid inlet valve and low-pressure switch.

3. Common Issues, Causes & Fixes

3.1 Purifier produces little or no water.

Possible causes:

- Clogged Stage 1 sediment filter.
- Low inlet pressure tripping the low-pressure switch.
- Failed booster pump.

Fix: Replace the sediment pre-filter if discolored. Confirm inlet pressure above 5 PSI. If the pump is silent when the tank is empty, test the pump for 24V at its terminals; replace if dead.

3.2 UV indicator is red / alarm beeping.

Possible causes:

- UV lamp at end of life (8000-hour rating).
- Flow sensor not detecting flow.

Fix: Replace the UV lamp and reset the hour counter. If the alarm persists with flow, inspect the flow sensor wiring to the ballast.

3.3 Water tastes off or has odor.

Possible causes:

- Exhausted carbon block (Stage 2).
- RO membrane fouled.

Fix: Replace the carbon block. If TDS out is above 50 ppm, replace the RO membrane and sanitize the tank.

4. Maintenance Schedule

Task	Interval	Notes
Replace sediment + carbon pre-filters	Every 6 months	Stages 1 and 2.
Replace RO membrane	Every 24 months	Sooner if feed TDS is high.
Replace UV lamp	Every 12 months	Rated 8000 hours.
Sanitize tank and lines	Every 6 months	Use food-grade sanitizer.